

런던 젠트리피케이션 연구

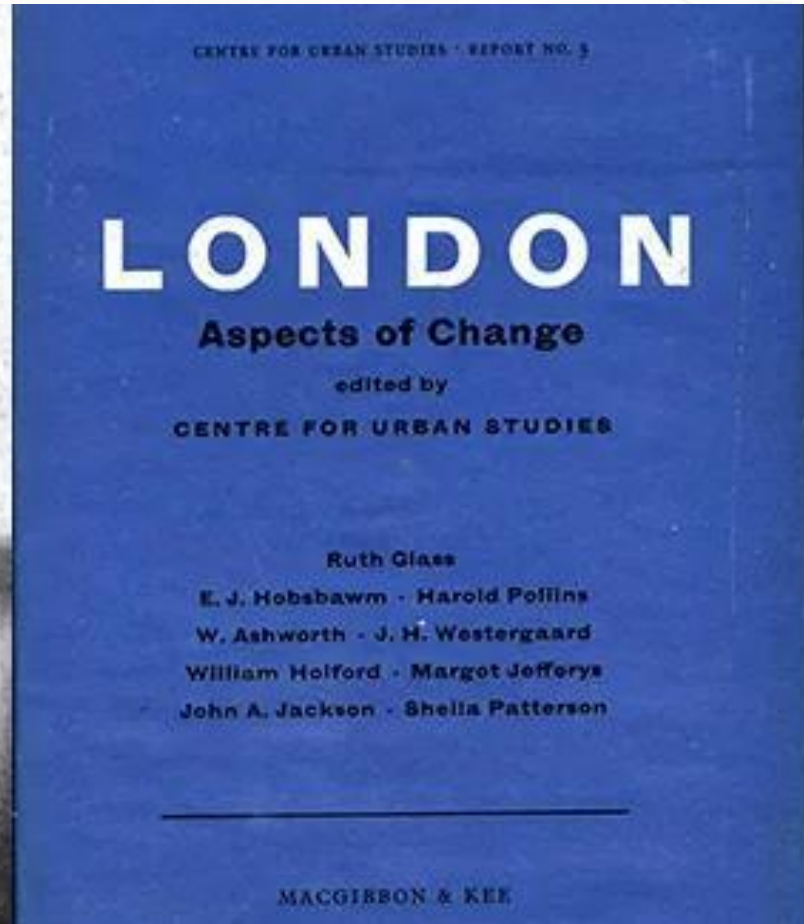
데이터마이닝 방법 및 실습 프로젝트 최종 발표
[팀 4] 박태현, 오종현, 이예찬, 박창현



Gentrification

젠트리피케이션

구도심이 번성해
중산층 이상의 사람이 몰리면서
임대료와 집값이 오르고
원주민이 내몰리는 현상



영국인 사회학자 루스 글래스(Ruth Glass)가
저서 **London: Aspects of change**에서 최초 제안



The long read

This article is more than 1 year old

How to reduce the damage done by gentrification

Olympic Legacy in East London: Displacement & Gentrification

By Felicity Clarke • January 25, 2013

[Clique aqui para Português](#)

Last year's London Olympics is widely viewed as a success. From Danny Boyle's spectacular opening and closing ceremonies to the smooth organizational running of the Games, the event seemed to represent a model hosting of



London Olympics: Residents 'betrayed' over housing promise

20 July 2022

Share Save

Aurelia Foster
BBC News



The Queen Elizabeth Olympic Park

Only about 200 of the new homes built on the Olympic Park site are on offer at the cheapest levels of rent

NEWS | LONDON

Revealed: The London neighbourhoods 'gentrifying' most rapidly

Areas with biggest jumps in average incomes have also seen disproportionate decreases in the number of black residents, analysis shows

목차 및 담당 팀원

선행연구 검토 및 측정 기준
데이터 구조 확인 및 전처리 전반
임대와 매매 간 관계 규명 및 처리
분류 및 시각화
감지 성과 확인
지역 특성 확인 (Global)
지역 특성 확인 (Local)
측정 기준 세분화 제안
결론 및 한계점

이에찬
박창현
박태현
이에찬
박태현
오종현
박창현
오종현

Where Do They Go? The Destinations of Residents Moving from Gentrifying Neighborhoods

Lance Freeman¹ ,
Jackelyn Hwang², Tyler Hauptert³,
and Iris Zhang²

Urban Affairs Review
2024, Vol. 60(1) 304–348

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Freeman, L., Hwang, J., Hauptert, T., & Zhang, I. (2023). Where Do They Go? The Destinations of Residents Moving from Gentrifying Neighborhoods. *Urban Affairs Review*, 60(1), 304–348. <https://doi.org/10.1177/10780874231169921> (Original work published 2024)

Hwang's Measure

1. Gentrifiable

도심 지역에 위치 (and)

해당 지역 가구 소득 중앙값 > 도시 전체 가구 소득 중앙값

2. Gentrifying ②

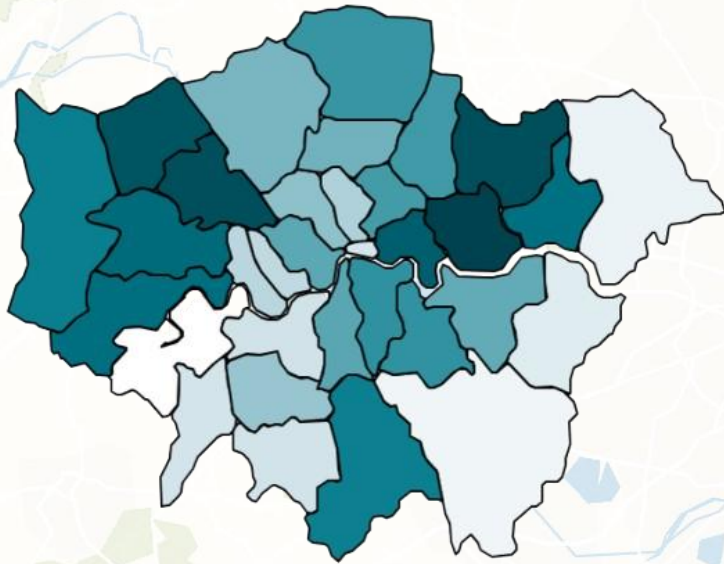
해당 지역 **대졸자 비율 증가율** > 도시 증가율 (or)
해당 지역 **가구 소득 중앙값 증가율** > 도시 증가율 중앙값

2. Gentrifying ②

해당 지역 주택 가격 중앙값 상승률 > 도시 상승률

Note: 가격은 평당 가격을 사용

Data 1: Census Datas



N = 4,994 (LSOA)
2011, 2021

Labour Market

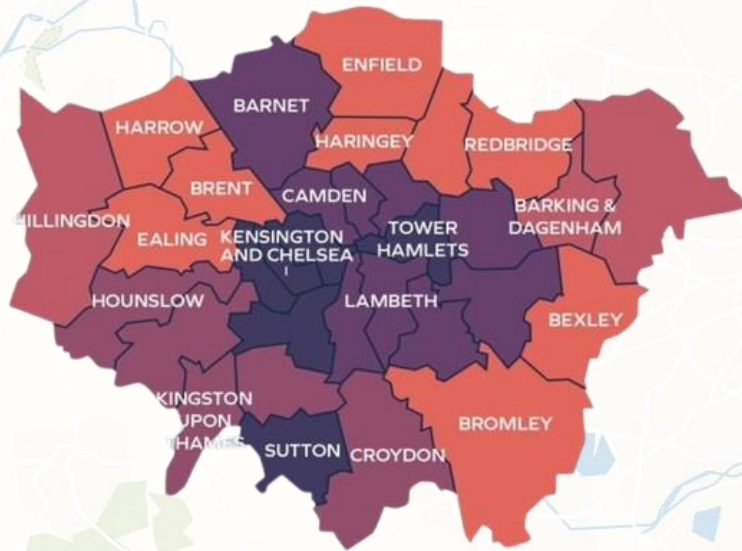
Demography & Migration

Ethnicity, Language & Identity

Housing

Qualifications, Health, Disability & Care

Data 2: London House Price Data



Geographical Variables

Property-related Variables

N = 418,201
1995-2024

Data 3-5: Additional Data

Postal Codes

Income

Qualification

**Geographical
Property-Related**

Mean, Median
by Local Authorities

Highest Qualification
by LSOAs

N = 330,907

N = 50
1999-2023

N = 4,995
2011, 2021

Data Wrangling

Problem

NA in Census Summary

NA in House Price Data
and Postal Codes

NA in Rent and Sale Price

Solution

모두 NA(or 0)인 LSOA 삭제
2011, 2021 요약 함수 다르게 정의

해당 매물이 속한
LSOA 내의 최빈값으로 대체

Rent Data, Sale Data로
나눠서 데이터 분석

Data Wrangling ②

```
csv_files = [  
    'master2011_rent.csv',  
    'master2011_sale.csv',  
    'master2021_rent.csv',  
    'master2021_sale.csv',  
    'gentrified_ward.csv'  
]  
  
dataframes = {file: pd.read_csv(file, engine='python') for file in csv_files}  
  
master2011_rent = dataframes['master2011_rent.csv']  
master2011_sale = dataframes['master2011_sale.csv']  
master2021_rent = dataframes['master2021_rent.csv']  
master2021_sale = dataframes['master2021_sale.csv']  
  
gentr2021_sale = master2021_sale.merge(dataframes['gentrified_ward.csv'], on = "ward code", how = "left")  
  
print(master2011_rent.shape, master2011_sale.shape, master2021_rent.shape, master2021_sale.shape)
```

✓ 1m 9.8s

(397661, 84) (397635, 88) (397661, 86) (397635, 90)

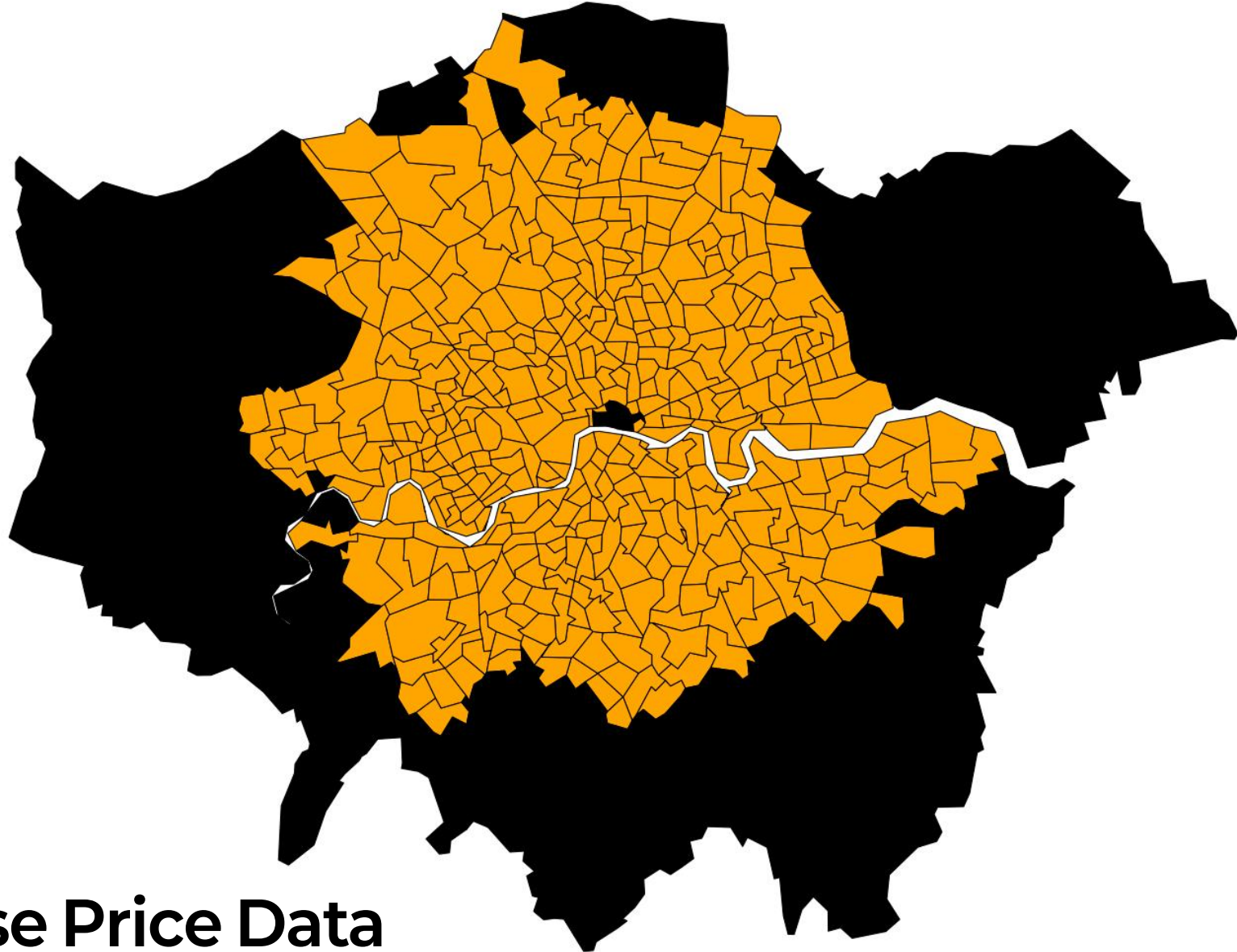
Unit of Analysis



Rent Price versus Sale Price

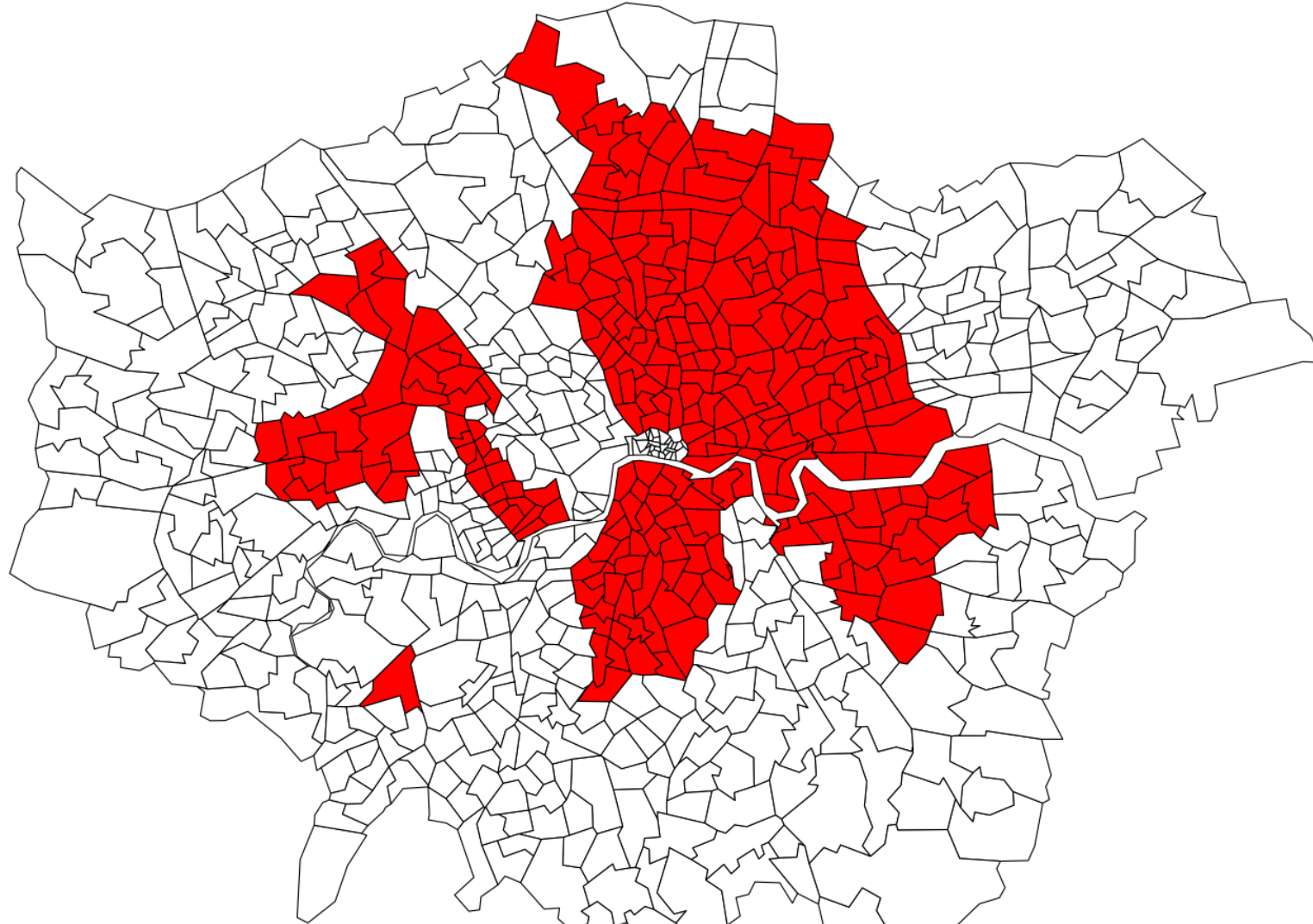


시각화: 도심 지역



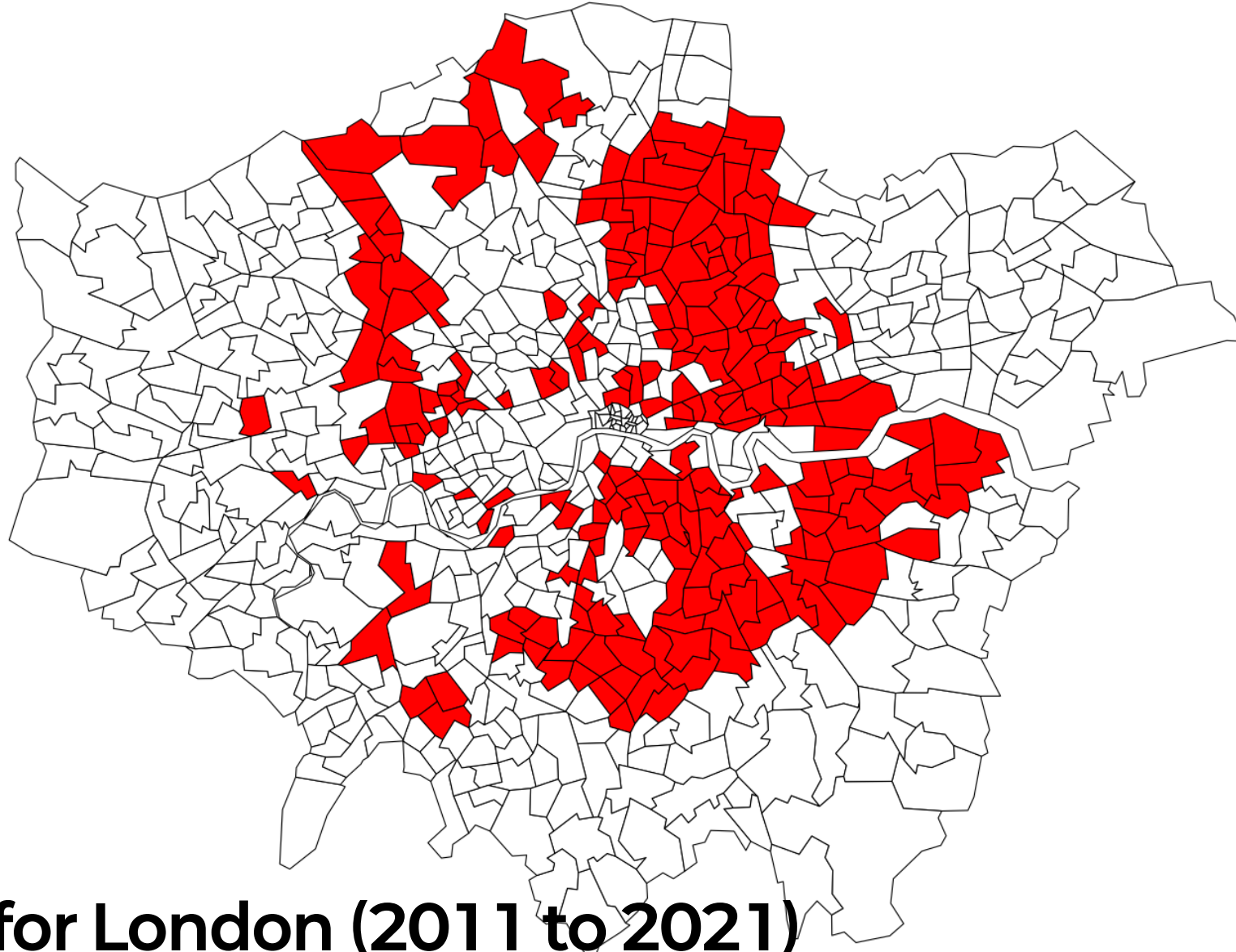
London House Price Data

시각화: 소득 기준 충족 지역



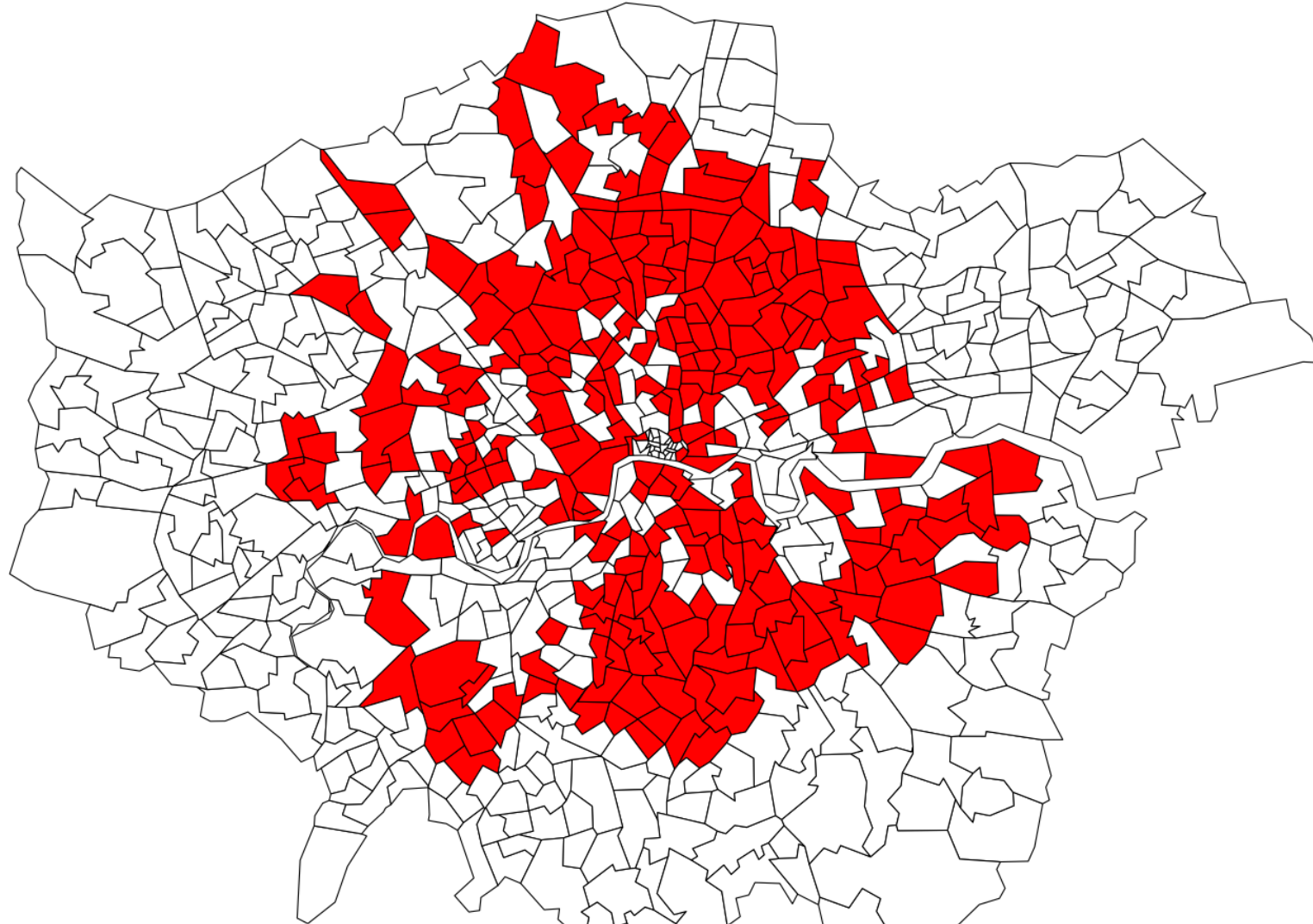
Income of Tax Payers Data (2011 to 2021)

시각화: 대졸자 비율 기준 충족 지역



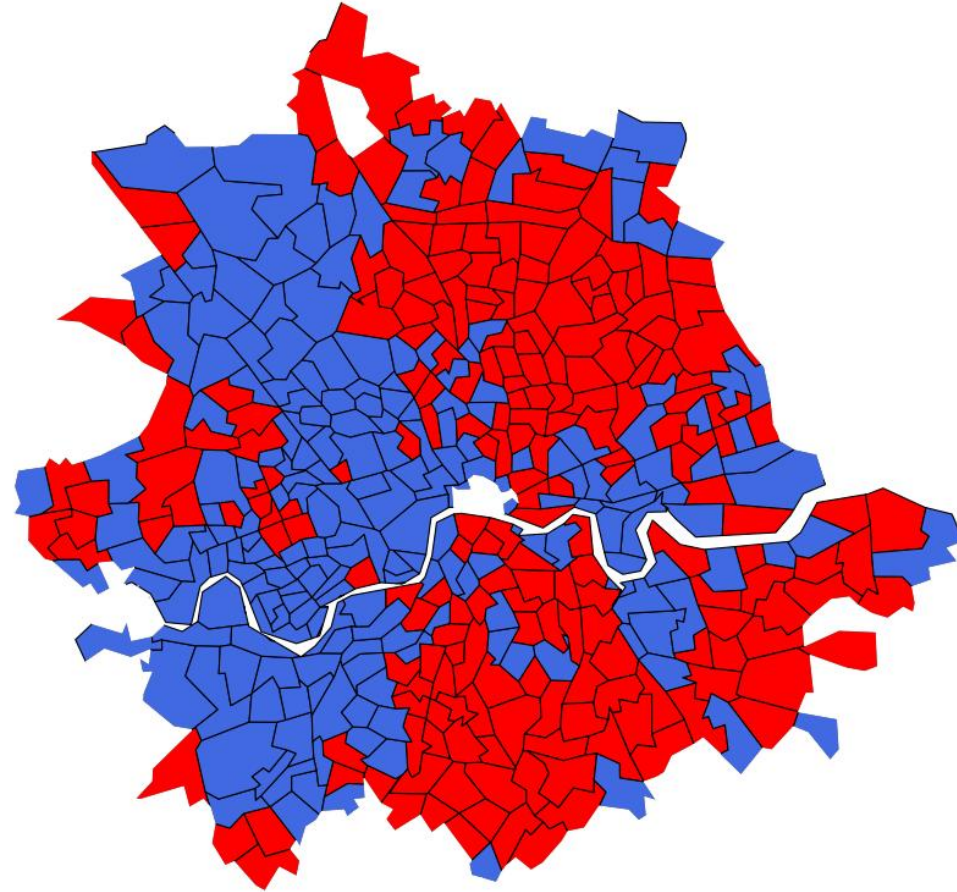
Census data for London (2011 to 2021)

시각화: 집값 상승률 기준 충족 지역



London House Price Data (2011 to 2021)

시각화: Hwang's Measure 최종 분류 결과

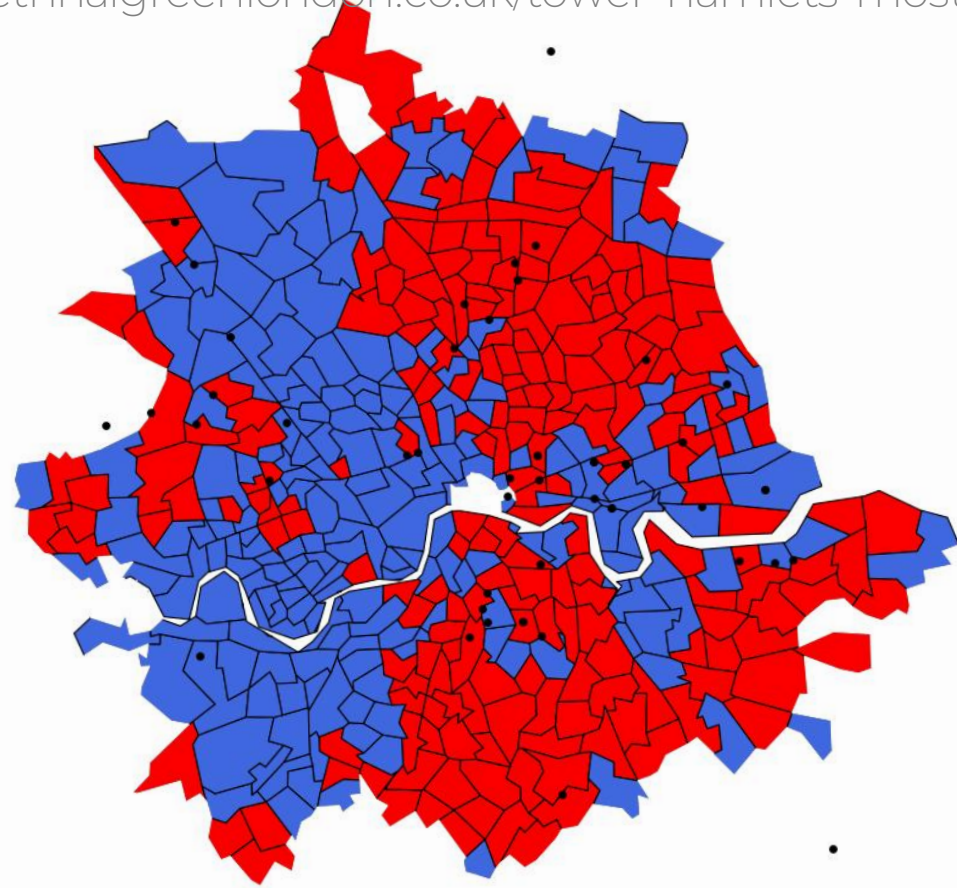


Gentrificated
Not Gentrificated

Hwang's Measure versus Reported

Reported

<https://bethnalgreenlondon.co.uk/tower-hamlets-most-gentrified-areas-in-london-2025/>

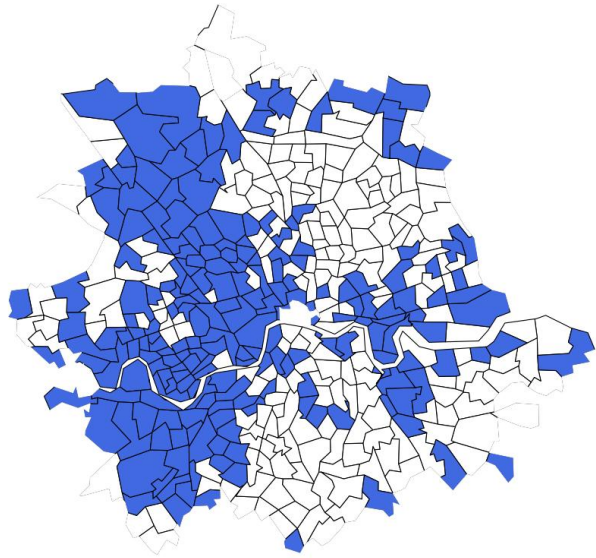


Contain Gentrified (31) → 79.4%
Otherwise (8)

Feature of **Non Gentrified** Wards

GBM Feature Importance for House Price

0 0.25



Professional Ratios

Higher managerial admin and professional most common

London Zone (지리적 접근성)

IMD Decile (지역 박탈지수)

Average Income

Distance to Station

Unpaid care most common

Altitude

Most common type of households are Owned Outright

Most common type is 4 or more bedrooms

Feature of **Gentrified** Wards

GBM Feature Importance for House Price

0 0.25



Average Income

Professional Ratios

Higher managerial admin and professional most common

London Zone (지리적 접근성)

Distance to Station

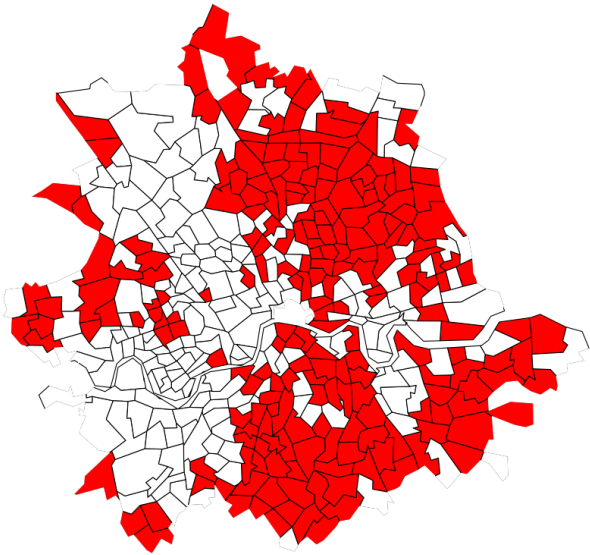
Altitude

Most common type is 4 or more bedrooms

Unpaid care most common

Most common type of households are Owned Outright

IMD Decile (지역 박탈지수)



Feature Comparison

GBM Feature Importance for House Price

Commonly important

Relatively Important in
Gentrified Region

Relatively Important in
Non Gentrified Region

Professional Ratios

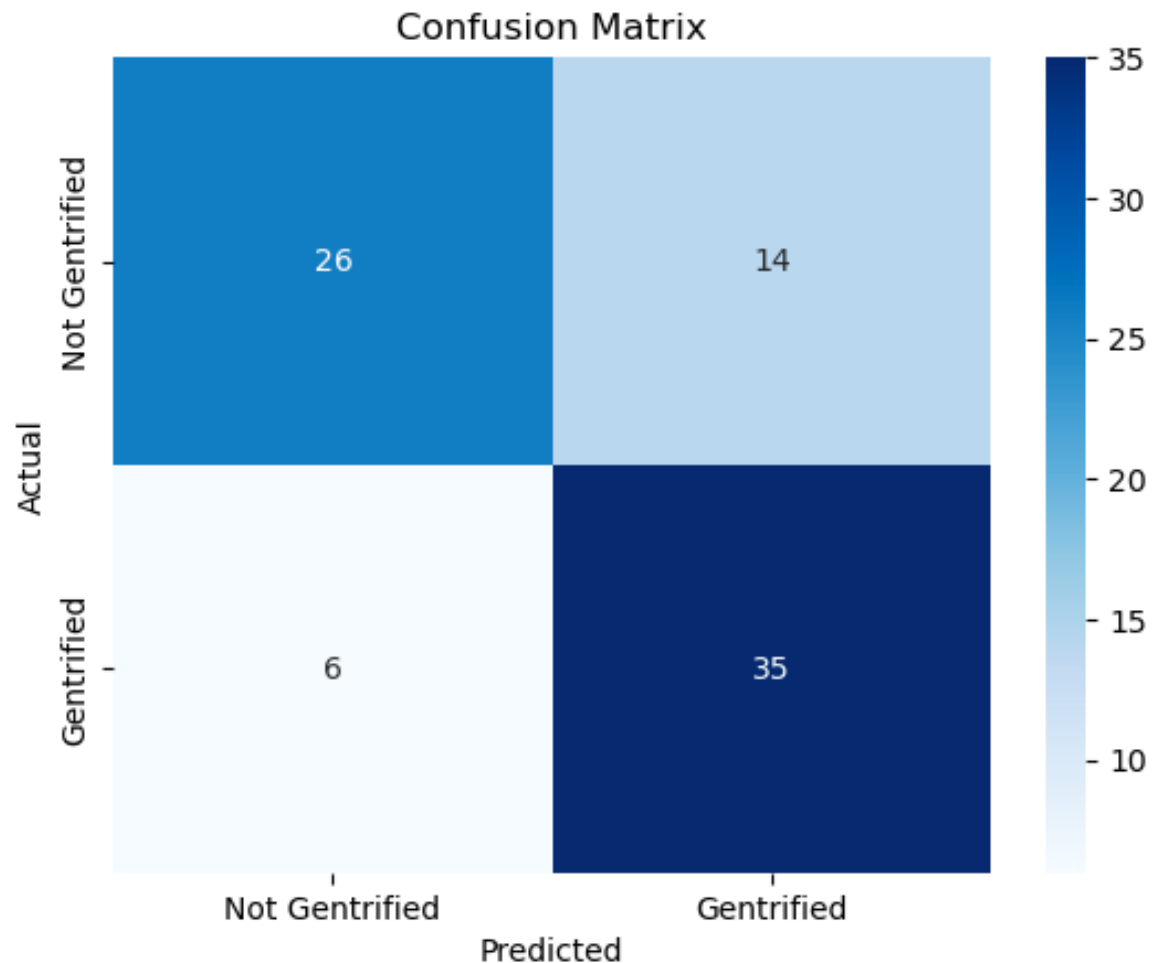
Higher managerial admin and professional most common

Average Income

London Zone (지리적 접근성)

IMD Decile (지역 박탈지수)

Classifier: Lasso Logistic Regression



LASSO Logistic Regression

Higher managerial and
Professional most common

Lower managerial and
Professional most common

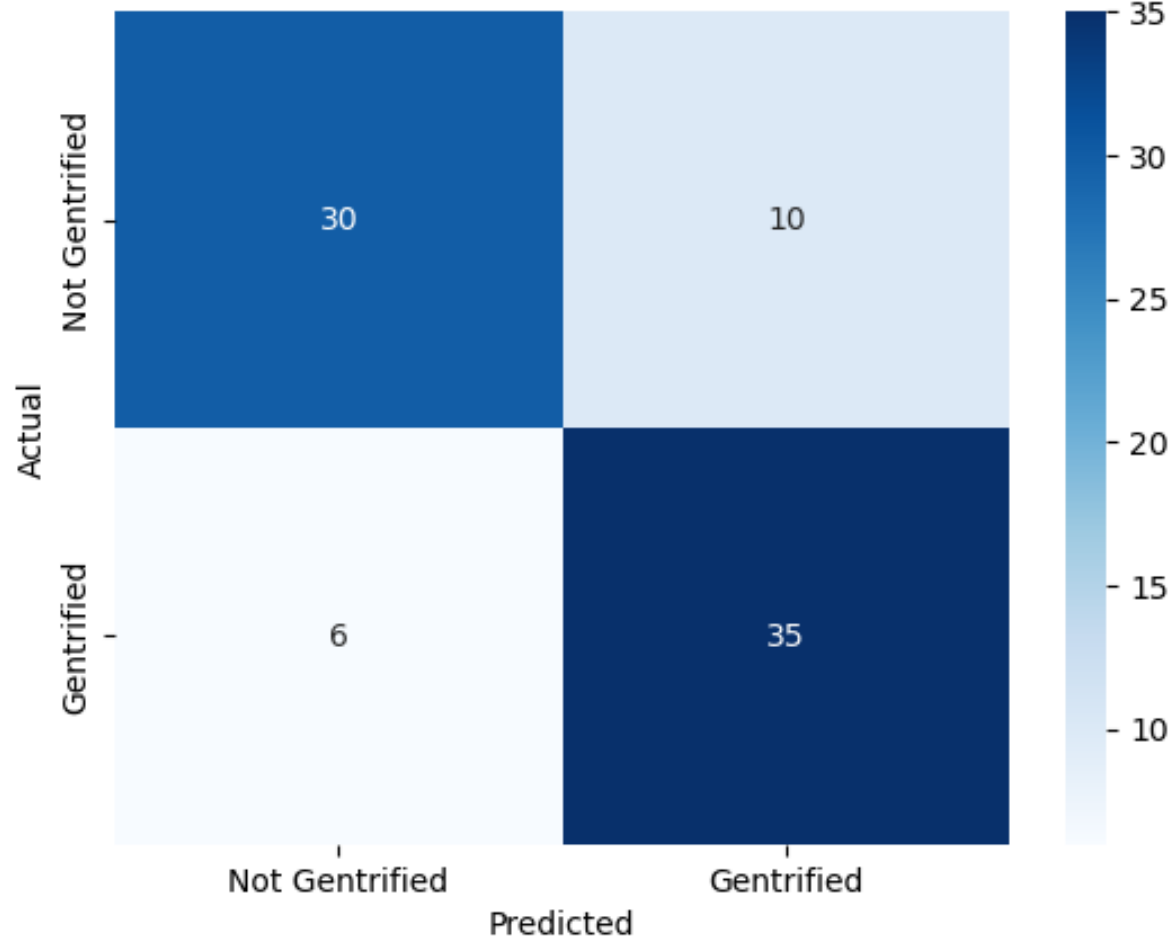
IDM Declie

Same ethic group
most common

0 ← Abs Coefficient → 0.35

Classifier: Random Forest

Confusion Matrix - Random Forest



Random Forest

Higher managerial and Professional most common

IDM Declie

Sale Estimate of Time2

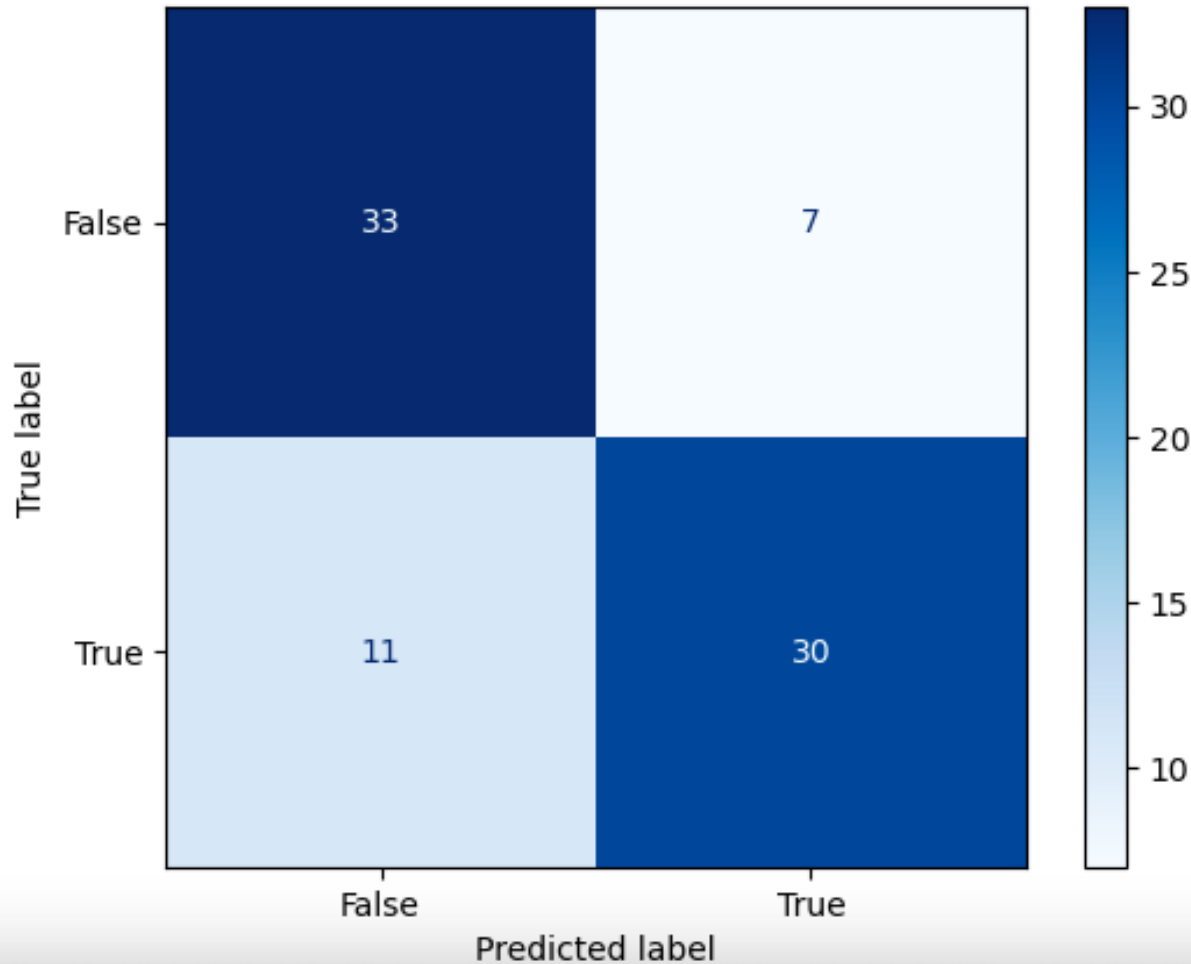
Average Income

Lower managerial and Professional most common

0 ← Importance → 0.14

Classifier: XGBoost

Confusion Matrix



XGBoost

Higher managerial and Professional most common

Lower managerial and Professional most common

Purpose built flat Most common

Same ethnic group most common

London Zone

0 ← Importance → 0.35

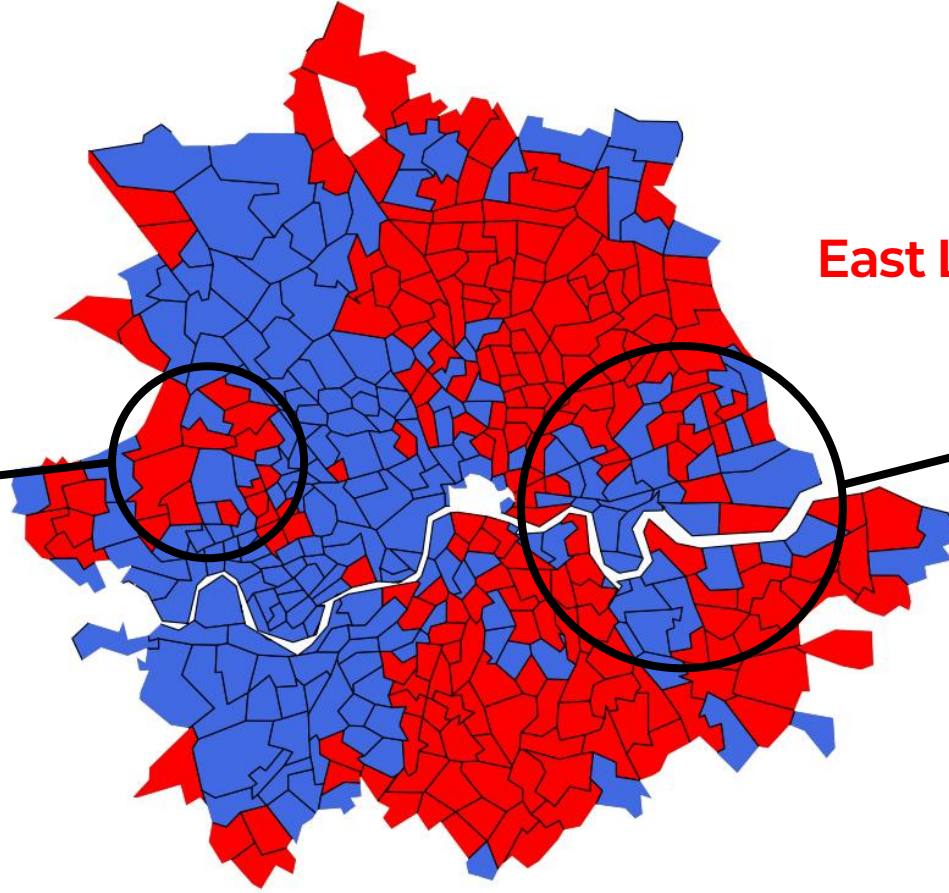
Classifier Comparision (Performance)

Metric	L1 Logistic	Random Forest	XGBoost
Accuracy	0.75	0.80	0.7778
AUC-ROC	0.7860	0.8506	0.8274
Precision (True)	0.71	0.78	0.81
Recall (True)	0.85	0.85	0.73
Precision (False)	0.81	0.83	0.82
Recall (False)	0.65	0.75	0.75

Local Difference

West London tends to be
Not Gentrified

Why?

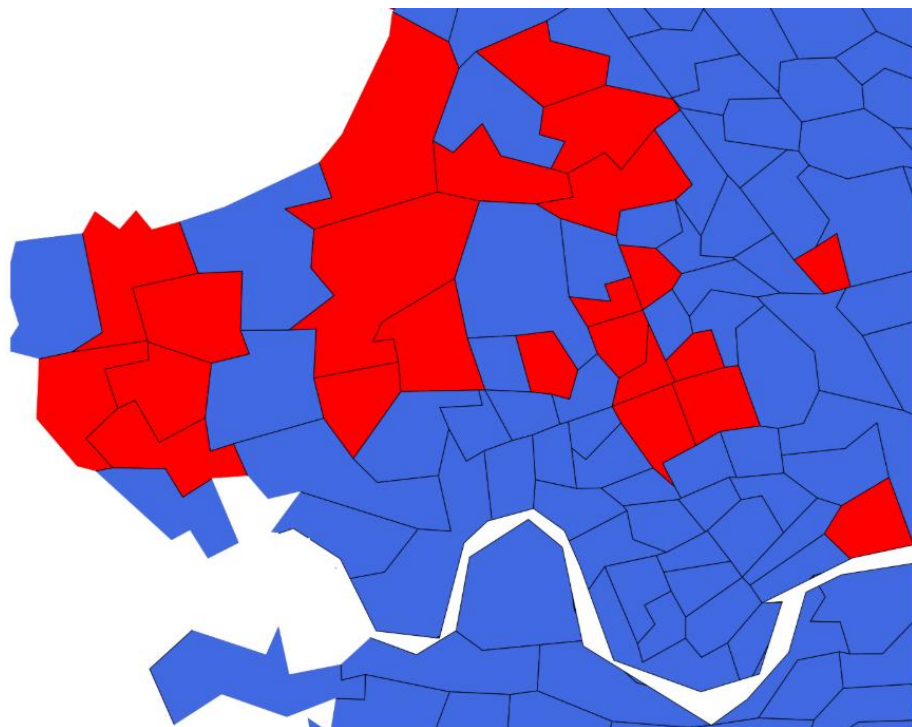


East London tends to be
Gentrified

Why?

Local Difference (West London)

중+장년, 하위직, 전통적 주택 위주



고위직, 1인 가구, 청년, 최신식 주택 위주

Gentrification 지역의 상대적 특징

주거 형태

Terraced + Covered Flat (+11.55%)
전통적 주택 형태 비율 높음

연령

30-50대 (+11.55%) 20대 (-17.26%)
중+장년층 비율 높음, 청년 비율 낮음

Deprivation

1, 2 Dimension Deprivation (+8.42%)
저소득층 비율 높음

1인 가구

1인 가구 비율 (-29.6%)
1인 가구 비율 매우 낮음

사회 계층/직업

고위직 (-38.6%) 하위직 (+31.67%)
직업 계층 낮음

Local Difference (East London)

Gentrification 지역의 상대적 특징

난방 방식

Gas (+7.9%), Electric (-8.2%)
신축 주택 비율 높음

인종

White British (+11.81%)
영국 백인 비율 높음

사회 계층/직업

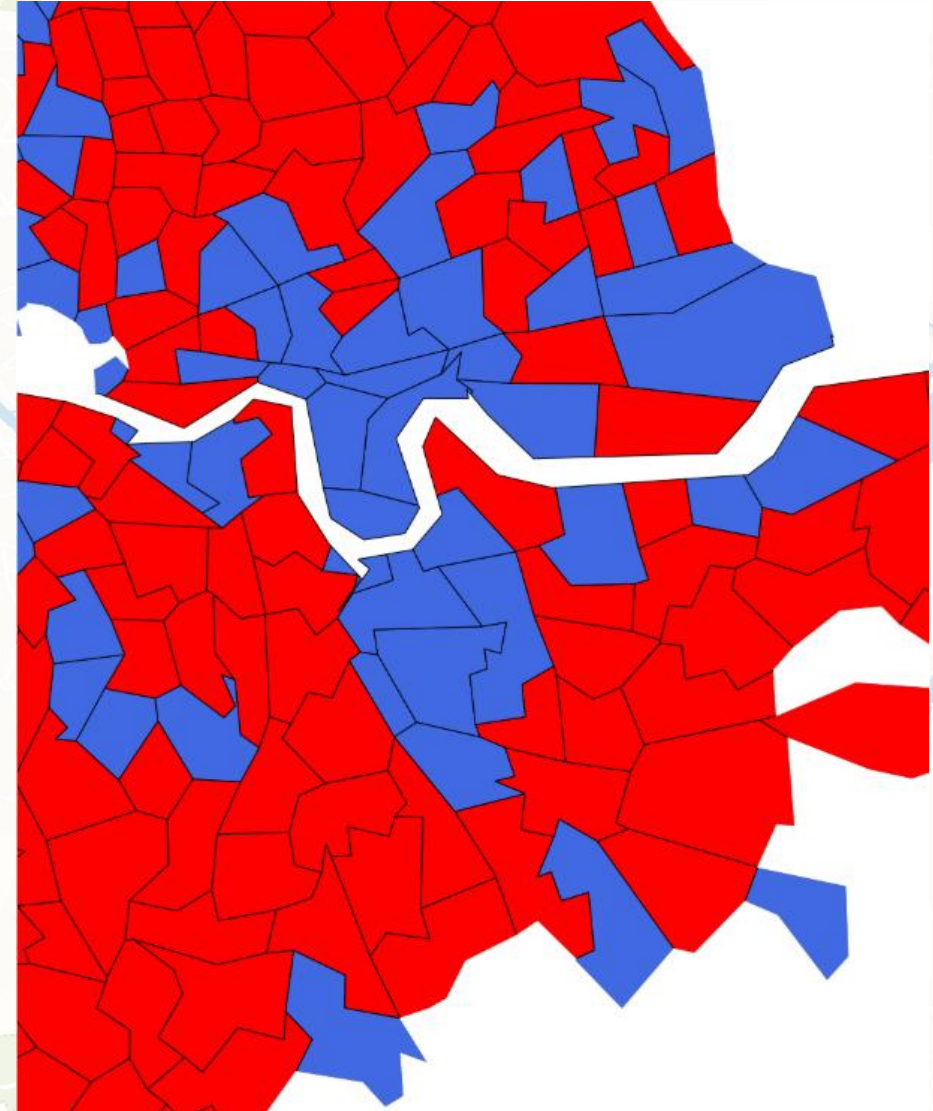
고위직 (-31.8%), 하위직 (+41.81%)
직업 계층 낮음

근무 지역

집 근처에서 근무 (+7.9%), 차량 없음 (+7.9%)
차량 없이 근무 용이한 지역

소유 형태

임대 (-15.2%), 자가 소유 (+12.5%)
자가 소유 비율 높음



Conclusions

Restructuring and instability of **Middle-Class spaces**

Socioeconomic drivers over locational factors

Divergent patterns across East and West London

Predictability driven by **occupational composition**

Limitations

Displacement **Tracking** Limitation

Median-Based Threshold Issue

Missing Contextual **Urban Development** Data

Lack of Temporal **Continuity**

References

[3pg] Glass, Ruth. London: Aspects of Change. MacGibbon & Kee, 1964.

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